

Knowledge, Attitude and Practises about Healthy Eating of Aspiring Coaches in Various Sport Disciplines.

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ABSTRACT

BACKGROUND: A coach is often considered as God by the athlete. A coach nurtures the athlete's talent and, helps them to develop to their full ability. A coach has many roles to play in an athlete's life like counsellor, mentor, nutrition advisor, facilitator etc. The right kind of knowledge is required to advise the athletes scientifically. Sports nutrition helps athletes to enhance performance and recover better. For better athletic output, nutrition and training should go hand in glove.

OBJECTIVE: To assess the Knowledge, Attitude and Practises of the Coaches about healthy eating and importance of nutrition in sports.

METHODS: The study was carried out at Sports Authority of India, Netaji Subhas Southern Centre, Bangalore. A total of 81 aspiring coaches were selected. Out of which 54 were male and 27 were female. A questionnaire method was used.

RESULTS: 52% of the subjects represented team sports like kabaddi, hHockey where as 39% of them were of individual events like badminton, swimming etc. The mean height among of the subjects was 169cm and mean weight was 66.8Kg. The mean BMI was 23.29, which falls under the normal category as all these were sportsmaen. About 38% of the subjects were in the particular sport for more than 15 years and majority of (the about 46%) them were in the sportsfield between 10 – 15 years. About 91% from individual sport and 96% of team sports had right knowledge of related to Carbohydrates. There was poor knowledge amongst the subjects in regard to protein. 98% of sportsmen has positive attitude about lack of iron causing fatigue. Awareness about carbohydrate loading was seen in 60% of the selected athlete subjects.

CONCLUSION: The overall scores indicated that most sportsmen had satisfactory knowledge of healthy eating habits,. Eexcept for protein consumption pattern and its requirement. Sports discipline did not have an influence on the nutritional knowledge of the sports mean.

INTRODUCTION:

According to Wikipedia(1) in sports, a coach is a person involved in the direction, instruction and training of the operations of a sports team or of

individual sports person. A coach may also be a teacher. A coach is often considered as Ggod by the athlete. A coach to an athlete god himself. A coach nurtures the athlete's talent, helps the

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athletes in developing to their full ability. A coach has many roles to play in an athlete's life like counsellor, mentor, nutrition advisor, facilitator etc. The right kind of knowledge is required by the coaches to advise the athletes scientifically. Sports nutrition helps athletes to enhance performance and recover better and faster. For better athletic output nutrition and training should go hand in glove. Nutrition plays an important role in sports performance because it helps an athlete to maintain ideal body weight, body composition specific to sports and faster recovery.(2) In India, After the commonwealth games, sports nutrition as a branch has gained importance. Sports nutrition as a field evolved predominantly after the Commonwealth games, 2010. A sports nutritionist advises the coaches and the athletes on the right nutrient practice and right nutrient timing of the foods to maximize the benefit of the foods from them. Athletes competing in team sports require them to switch intermittently between maximal-effort and low-intensity exercise, potentially causing large losses of body water, therefore, hydration is an important but often overlooked aspect of proper training.(3) The delay in production and accumulation of lactic acid in the athlete's body can enhance his performance. Enhanced performance can be result of delayed the Lactic acid production Adequate nutrition enhances aerobic capacity or Vo_{2max} ⁽⁴⁾.

Athletes often choose supplements

over adequate nutrition due to lack of knowledge or peer pressure. Most of the time they fall prey for the supplements, without realising if that does any good to them or not. Ever increasing pressure on athletes and bodybuilders to perform and do their best along with increasing competitiveness of sports pushes many athletes to add pharmacological agents and dietary supplements to their training and nutritional eating plans to improve their performance levels.(5)

A sound and scientific basic knowledge is very essential for the coaches to advise their athletes better. The intermediate level of KAP (Knowledge, Attitude and Practice), organized and regular education courses are highly recommended along with paying more attention to the curriculum taught.(6) A proper knowledge, attitude and practice (KAP) of sports nutrition is associated with many positive outcomes for athletes including optimal health and performance.(7) (won't be able to change as it's a reference.)

However, only a few studies assessing nutritional knowledge, attitudes and practices (KAP) are available, which presents the need of the study.

OBJECTIVE:

1. To assess the Knowledge, Attitude and Practices of the Aspiring Coaches about healthy eating and importance of nutrition in sports..

METHODS: The study was carried out at Sports Authority of India, Netaji Subhas Southern Centre, Bangalore. A total of 81 aspiring coaches were selected

through purposive selection. Out of which 54 were male and 27 were female. A questionnaire method was used.

The Aspiring Coaches were taught basics of Sports nutrition understand the right concepts of nutrition like carbohydrates, proteins fats, hydration etc. They were also given basic inputs of foods for each sport discipline, or foods pre competition, during competition and post competition.

A standardised well framed and validated questionnaire was developed which comprised of question related to the nutritional knowledge and attitude and practices. A pilot study was run before all the data were collected. It was followed by an interview method. A similar study on nutritional knowledge of athletes by Mitchell Face(8) was referred to develop the questionnaire.

PROCEDURES:

Questionnaires were distributed to the athletes with a covering letter mentioning the need of the study, the purpose of the study. A paragraph about confidentiality was also mentioned in the covering letter. A consent form in both English and regional language was taken from the aspiring coaches after explaining about the study. The primary investigator collected the questionnaire there after. The study was conducted after the sports nutrition academic classes were conducted.

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basic inputs of foods for each sport discipline, or foods pre competition, during competition and post competition during the academic classes.

RESULTS AND DISCUSSION

DEMOGRAPHIC INFORMATION:

A total of 81 aspiring coaches were selected. Out of which 66%(n=54) were male and 44%(n=27) were female.(Table 1). The age group of the participants was between 23 years to 37 years. The subjects between the age group 23 years to 37 years. Among the subjects 64% (n=52) were unmarried and 46%(n=46) were married. Individual sport data was collected. A total of 9 sport disciplines were studied.(Table 2). It was further divided into individual and team sports. About 52%(n=42) of the subjects represented team sports like kabaddi, hHockey where as 39% (n=39) of them were of individual events like badminton, swimming etc.(Figure 1).

The mean height among of the subjects was 169 cm and mean weight was 66.8Kg. The mean BMI was 23.29, which falls under the normal category as all these were sportsman. The duration of the participants practising in their particular sport was studied. Almost half of the participants 46%(n=46) were practising the sport between since 10 – 15 years. We also had Around participants 38% (n=31) of participants who were practising practiced that particular sport for more than 15 years. About 38% (n=31) of the subjects were in the particular sport for more than 15 years and majority of the about 46%(n=46) were in the sport between 10 – 15

years.(Figure 2).

NUTRITION KNOWLEDGE:

Enhancing nutrition knowledge and improving dietary intakes of the athletes is the main aim of nutrition education.(9)

The nutrition knowledge of the aspiring coaches was studied through a set of questions based on their knowledge in the questionnaire. The answers were scored with a help of a scoring system. The nutrition knowledge of the aspiring coaches was good. When questioned and co related the scores were compared, they showed poor knowledge about protein. Aspiring Coaches were generally knowledgeable with regard to nutrition.The common fad the have is about protein which was evident in the study. Protein consumption is always believed to be the most important part of the diet by athletes especially the once who are looking at body building. Also when it comes to the quantity consumed by any individual, there are too many myths attached to it. They were also aware about the effects of dehydration.

About 91% from individual sport and 96% of team sports had right knowledge of Carbohydrates and its sources. The poor knowledge amongst the subjects was in regard to protein. 36% from individual sport and 26% from team sport had right knowledge on protein and its requirement.100% from both sports knew the kind of calories obtained by fat. 82% from individual sport and 87% of team sports had good knowledge on vitamins and minerals.

96% from individual sport and 97% of team sports had adequate knowledge on hydration, effects of dehydration and fluid balance.(Figure3).

ATTITUDE RESPONSES:

Extreme eating behaviour and such attitude in the athletes can lead to the feeling of loss of control by the athletes.(10)

About 100% from individual sport and 98% from team sports had positive attitude about lack of iron causing fatigue. However iron deficiency anaemia is one of the common nutritional problems seen by most of the athletes, especially girls.92% from individual sport and 100% from team sports agreed that nutritional needs of athletes is different from normal population. 23% from individual sport and 21% from team sports had positive attitude for vitamin supplementation. When asked about important foods for enhancing performances, participants in team sports 52% of participants in team sports had right attitude towards consuming them. Also 41% of participants from individual sports showed right responses for these foods.41% from individual sport and 52% from team sports had right attitude to about ghee, almonds and milk enhancing performance.92% from individual sport and 100% from team sports had positive attitude about dietary changes required with change in climatic conditions and seasons.(Figure4).

PRACTICE OF FOODS: FOOD CONSUMPTION PRACTISES:

Reviewing the diet of Olympic athletes as a concept existed in the past

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centuries. In support of. Lack of qualitative description of their diets is there are only and only handful of data is present. (11)

A Right eating practice of foods were as seen generally in both individual and team sports. (Table 3). Both the sport group and aspiring coaches had similar response to the questions asked.

About 87% from individual sport and 90% from team sports changed their dietary pattern during the time of competition. 67% from individual sport and 72% from team sports did not skip meals prior to competition. The aspiring coaches were not well informed on the good food choices to be had consumed pre-competition but . They managed to eat something before the competition. Though they were not well informed of pre competition meals they managed to eat something before the competition. 41% from individual sport and 31% from team sports were practising the usage of sports drinks during training phase. About 61% from individual sport and 60% from team sports practised carbohydrate loading prior competitions whereas nearly half of them did not practice carbohydrate loading either with lack of knowledge about it or due to the fear of weight gain. A very minimum nominal percentage of 36% from individual sport and 28% from team sports used to eat energy bars during the training. The concept of energy gels was very new to the aspiring coaches. Though they were in the sport for so long they revealed that the concept was new to them. They

were not aware of the right usage, timing and dosage of the energy gels. 18% from individual sport and 15% from team sports used energy gels occasionally. Eating banana for energy or for protein is as commonly believed as a practice from ages. Banana consumption as pre competition snack and during the competition was a common practice. When the aspiring coaches were playing their involved in competitive sports, then also they had a similar practice. The aspiring coaches ate bananas before the training and competition even when they played the sport earlier. About 72% from individual sport and 74% from team sports

had banana regularly during training. The rest of the participants also believed in weight gain on that eating banana. would cause weight gain so they also did not have the practice of eating a fruit before competition. The rest half who did not eat the fruit answered that banana causes weight gain hence the avoidance. Isotonic sports drinks are very commonly available. The commercial available sports drinks are available in different brands and flavours. They come in different brands and flavours. Though isotonic drink can also be considered as a supplement aspiring coaches were not aware. hardly they would agree on that. Nearly half, 54% from individual sport and 55% from team sports consumed isotonic drink. Only a few participants had the knowledge about it. Other participants consumed sports drinks being influenced by the other coplayers because of influence of peer pressures as they believed it can enhance

performance. Rest all consumed due to peer pressure for performance enhancement as they believed.

CONCLUSION:

Nutritional intake plays a very important role in enhancing an athlete's athletic performance, and their ability of physical and mental performance. Optimum dietary intakes are often compromised due to travelling, lack of knowledge and demanding training. (12)

The aspiring coaches were provided with academically trained knowledge on sports nutrition. They gained sufficient knowledge. The scores also indicated that they gained sufficient knowledge the same.

The overall scores indicated that most aspiring coaches had satisfactory knowledge of healthy eating habits, e. Except for protein consumption, eating pattern and its requirement.

The study also indicated the need for nutrition education as a primary need for the coaches to understand and help the athletes in enhancement of their performance enhancement. Right nutritional intervention at right time can help athletes achieve what they aim for. Also There were sport specific variation in eating practices consumption of foods or and some had practices or even

food fadfadisms due to the influence of peers and , coaches. whoSome of them believed in something particular or even due to the traditional practices of the sport itself. It is need of the hour to educated the aspiring coaches, as the carry the legacy forward of scientific training and appropriate nutrient timing. Indian food dairy is a huge one. EIn India each region has its own regional foods. Athletes needs to be introduced to staple diets of the other states, as they might travel to different placesstates for the competition. The aspiring coaches' knowledge on different regional foods makes it easy for them for to guide their athletes during travel and competition.

India being a vast country acclimatization of different foods have to be thought of as the athletes might attend the competition in any part of India. Also the aspiring coaches can be educated with basic knowledge of substitutes available so that they can help their athletes during travel and competitions. Food fadsdism like prohibited foods can have an impact on the performance of the athletes. Hence it is important to educate the aspiring coaches on right practicse. Basic nutrition education for the aspiring coaches is coaches are most important. It helps to improve the Knowledge, Attitude and Practice regarding healthy eating.

List of Tables and Figures

Gender	Number of subjects(n)	Total percentage(%)
Male	54	66%
Female	27	44%

Table 2 : Number of participants from each sports discipline (n,%)

Sport Discipline	Number of Subjects(n)	Number in percentage (%)
Athletics	17	21
Badminton	10	12
Hockey	12	15
Kabaddi	14	17
Kho-Kho	8	10
Softball	2	3
Swimming	8	10
Taekwondo	4	5
Volleyball	6	7

Table 2 : Food practise pattern of subjects (%)

SI No	Questions related to Practise	Yes(%)		No(%)	
		Individual Sport	Team Sport	Individual Sport	Team Sport
1	Does your dietary pattern change at the time of competition?	87	90	13	10
2	Do you skipping meals prior to competition?	33	28	67	72
3	Do you consuming sports drinks during practice?	41	31	59	69
4	Do you practice carbohydrate loading prior to competition?	61	60	39	40
5	Are you having the habit of taking energy bar during training?	36	28	64	72
6	Do you consume energy gel during training?	18	15	82	88
7	Will you practice of eating bananas during training?	72	74	28	26
8	Do you consume isotonic sports drink (6g/100ml) during training?	54	55	46	45

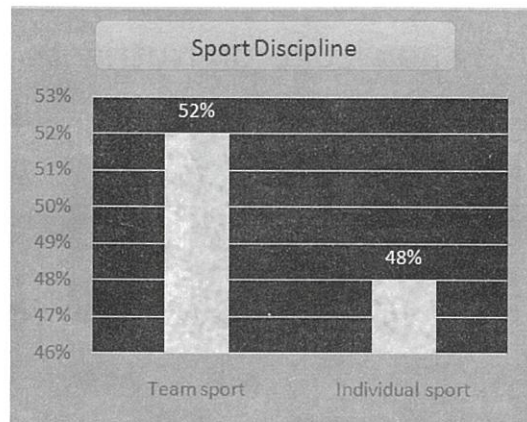
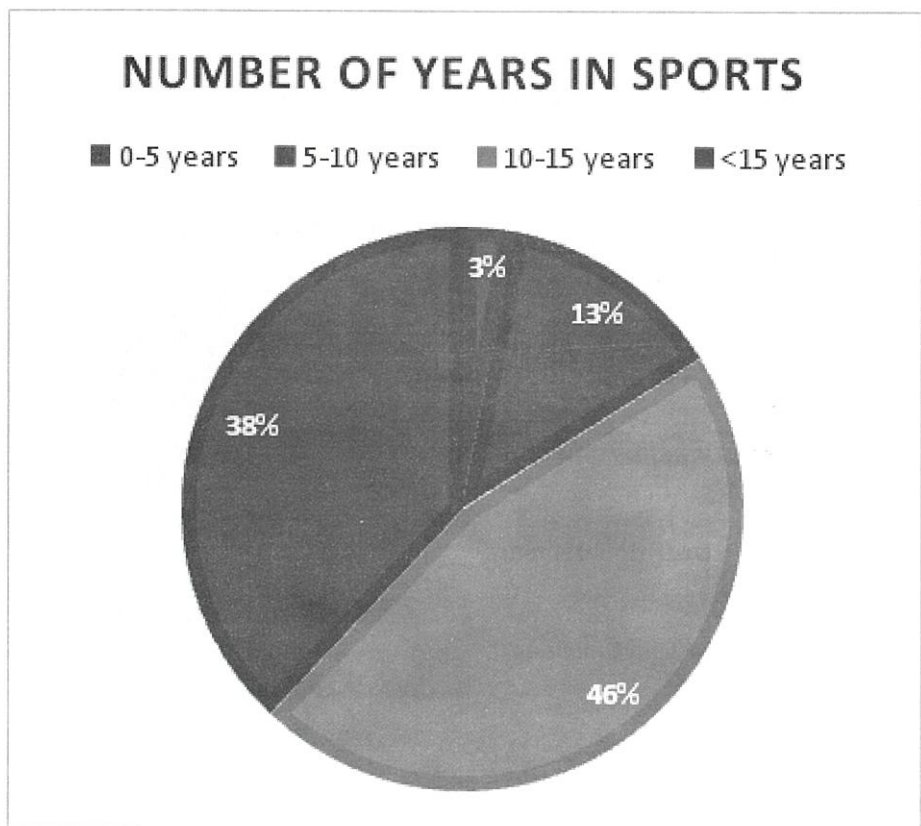
Figure 1 : Participants of team sports and individual sports (%)**Figure 2: Experience of subjects in sports (%)**

figure 3: Knowledge of nutrients among subjects (%)

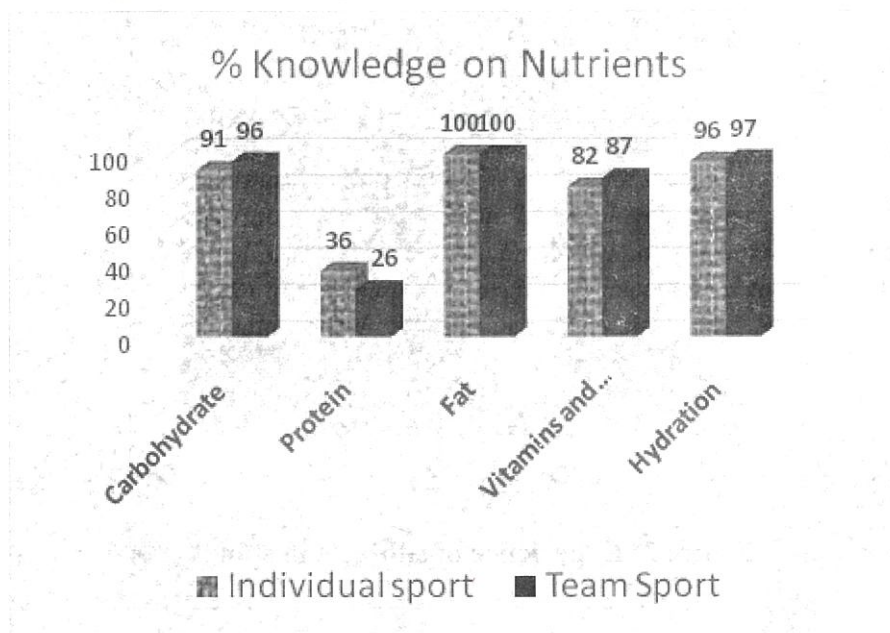
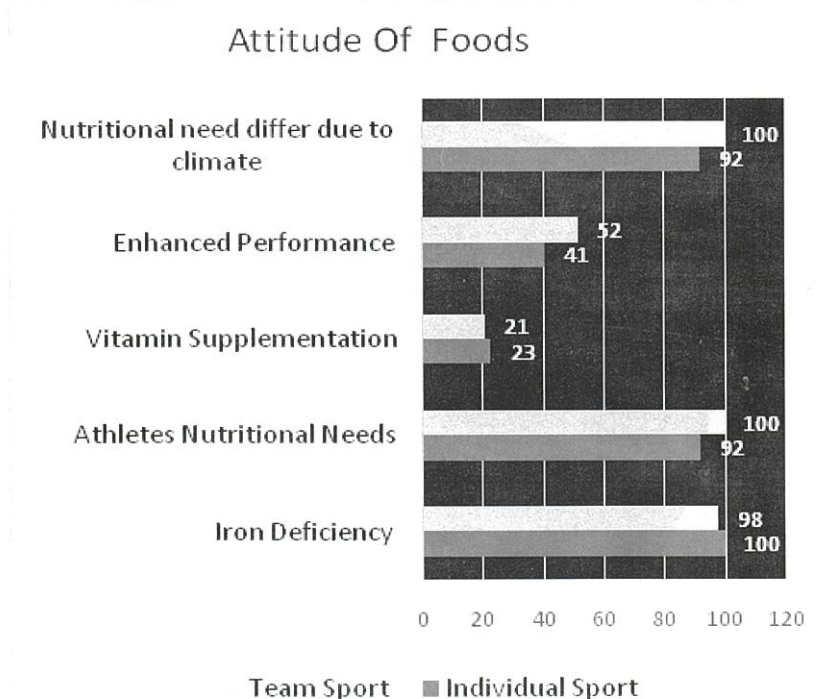


Figure 4: Attitude of subjects towards food (%)



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